



Figure 2: Docker Labs and genAI

deployment of conversational agents for incorporation into websites, customer service, or embedded systems.

Here's an example of the integration of generative AI in a Docker container.

Set up a small Flask server using the Hugging Face Transformers library to generate text with GPT. For example, a prompt 'Once upon a time', sent to a local web server, will return the text generated by GPT-2.

Next, create your project folder by creating a new directory:

```
mkdir my-gpt2-docker-app
cd my-gpt2-docker-app
```

Now, create *app.py*:

```
# app.py
from flask import Flask, request, jsonify
from transformers import pipeline

app = Flask(__name__)
generator = pipeline("text-generation", model="gpt2")

@app.route("/generate", methods=["POST"])
def generate():
    mydata = request.get_json()
    myprompt = data.get("myprompt", "Hello world")
    myresult = generator(myprompt, max_length=50, num_return_
sequences=1)
    return jsonify(result)
```

The requirements.txt is:

```
flask
transformers
torch
```

The Dockerfile is:

```
# Use an official Python image
FROM python:3.10-slim

# Set working directory
workingdir /app

# Copy code
COPY . .

# Install dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Expose port
EXPOSE 5000

# Run the app
CMD ["python", "app.py"]

# Build the image
```

You can now build and run the Docker container. Open the terminal inside *my-gpt2-docker-app* directory and run: